

# Hanyuan Zhang

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## EDUCATION BACKGROUND

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|-----------------------------------------------------|-------------------|
| University of California San Diego (UCSD), USA      | 09/2025 - 07/2027 |
| MS Machine Learning and Data Science EC93           | GPA:3.67          |
| Xi'an Jiao Tong Liverpool University (XJTLU), China | 09/2021 - 07/2025 |
| BSc Information and Computing Science               | GPA:3.77          |

## Research Interesting

LLMs and machine learning for clinical reasoning; clinical trial prediction and patient matching; EHR and clinical text understanding; causal and predictive modeling with real-world clinical data; evaluation of medical AI systems.

## PUBLICATIONS (\* means co-first author)

- [1] Jianyou Wang\*, Youze Zheng\*, Longtian Bao\*, **Hanyuan Zhang\***, Qirui Zheng, Yuhan Chen, Yang Zhang, Matthew Feng, Maxim Khan, Aditya K Sehgal, Christopher D. Rosin, Mohan Paturi, Umber Dube, Leon Bergen. *CT Open: An Open-Access, Uncontaminated, Live Platform for the Open Challenge of Clinical Trial Outcome Prediction*. In *Conference on Language Modeling (COLM)*, 2026 [\[Link to paper\]](#)
- [2] Wen Sheng, Jun Zhao, Zhengdi Sima, Jiajun Liu, Han Lu, **Hanyuan Zhang**, Zhong Zheng, Zhihong Zhang, and Daoping Zhu. *Advancing Pharmacological Treatment Effectiveness with Dual-Encoder Model in Plain Scan Liver Tumors*. In *Indian Journal of Pharmaceutical Sciences*, 2024 [\[Link to paper\]](#)

## Manuscript in Preparation

- [3] Youze Zheng\*, Jianyou Wang\*, Yuhan Chen\*, Matthew Feng\*, Longtian Bao, **Hanyuan Zhang**, Maxim Khan, Aditya K Sehgal, Christopher D. Rosin, Umber Dube, Mohan Paturi. *DeepImagine: Clinical Trial Outcome Prediction via Stepwise Local Counterfactual Imaginations*. Archive 2026 [\[Link to paper\]](#)

## RESEARCH EXPERIENCES

**Laboratory for Emerging Intelligence** 10/2025-Present

Research Assistant, Advisor: prof. Ramamohan Paturi, Professor

**Scientific knowledge graphs: Research Map. 04/2026 - present**

- Project Foundation: Building on a collaborator-developed biomedical knowledge graph derived from 311K+ PMC review articles (covering 40M+ primary papers), comprising ~1.9M question-level research narratives and millions of LLM-verified edges.
- Research Contribution: Investigating graph-based methods to trace the evolution of biomedical research and exploring how these evolutionary trajectories can augment LLMs to forecast emerging research directions.

**CT Open: Live Clinical Trial Outcome Prediction Benchmark. (COLM 2026, co-first authors) [1]**

- Co-developed a live, contamination-resistant benchmark for predicting clinical-trial outcomes before public release.
- Built quarterly data curation, decontamination, verification, submission, and leaderboard pipelines.
- Released 7,292 trials/15,444 questions with 98%+ decontamination and 99% answer accuracy; accepted at COLM 2026.

**DeepImagine: Counterfactual Clinical Trial Outcome Prediction. (Archive 2026, Under review in NeurIPS 2026) [3]**

- Helped to contribute a stepwise method that adapts matched historical trials one experimental factor at a time.
- Helped to Evaluate seven-anchor counterfactual reasoning against RAG baselines for trial-outcome prediction.
- Improved GPT-5 macro-F1 from 72.4 to 80.8; manuscript under review at NeurIPS 2026.

**Zhu's Laboratory, XJTU**

01/2023-03/2024

Research Assistant, Advisor: prof. Xiaohui Zhu

**Non-Contrast Liver Tumor Segmentation.**[2]

- Conducted non-contrast 3D liver-tumor segmentation on 40 PSLT volumes comprising 10,923 CT slices.
- Implemented YNetr, a wavelet-guided dual-encoder 3D Transformer, in PyTorch/MONAI.
- Achieved 62.63% Dice, 1.22 points above the strongest baseline;

## **WORK EXPERIENCES**

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**Shandong Inspur Innovation and Entrepreneurship Technology Co., Ltd., Jinan, China**

06/2025 - 09/2025

*AI Intern*

- Built a ResNet18 item-recognition system for China Petroleum inventory operations.
- Developed BERT-based NER and relation-extraction models for a Qwen-assisted knowledge graph.
- Achieved 92% item-classification, 78% NER, and 75% relation-extraction accuracy.

**Suzhou Kaixide Information Technology Co., Ltd., Suzhou, China**

06/2023 - 08/2023

*Data Scientist Intern*

- Developed and deployed an education evaluation system for large-scale student-performance analysis.

**Shandong Health Big Data Management Center, Jinan, China**

01/2022 - 03/2022

*Health Data Scientist Intern*

- Using machine learning method to analyze real-world adolescent health and biometric data from Shandong Province

## **EXTRACURRICULAR EXPERIENCES**

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**The University of Hong Kong Summer Institute**

07/2024

- Completed hands-on coursework in computer vision, large language models, and embodied AI.

**Team Leader | CPI Forecasting with RNN/LSTM | 'Gauss Cup' Mathematical Modeling Challenge**

06/2022

- Led a three-member team to build a CPI forecasting pipeline from public historical data.
- Engineered time-series features and trained RNN/LSTM models with regularization and early stopping.
- Delivered visual analyses, model evaluation, and a report on societal impact.

## **AWARDS AND HONORS**

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- The 2nd Prize | "Gauss Cup" China Collegiate Mathematical Modeling Challenge

06/2022

## **SKILLS**

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**Programming Languages and Tool:** Java | Python | SQL | MATLAB | Neo4j | Excel | Word | PowerPoint

**Libraries:** NumPy | SciPy | Pandas | PyTorch | Scikit-Learn | Matplotlib

**Language:** Chinese (Native language) | English (Fluent, TOEFL scores:105)